Autonomous Visual Object Finding with an e-puck: Perception, Route Planning and Safety

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Abstract

[TODO: 4–6 sentences: task (e-puck must find the station showing the mission target and stop at its observation position within 4 min, no collision); approach in one line each for perception, route planning and safety; headline result (success rate and median time on the three student missions and the 96-run Monte Carlo benchmark).]

1. Introduction

The task is simulated in Webots [4] with an e-puck robot [5]. [TODO: The problem: an e-puck in a known arena with eight observation stations; one station displays the mission target. Success = stopping within 0.20 m of the correct observation position, no collision, within 4:00 simulation time.]

[TODO: What makes it hard: target identity is only available visually (no world-file or texture access); low camera sees only the lower half of each poster; several stations are in shadow; robot arrives with heading error; narrow corridors.]

[TODO: Project idea in one paragraph: map-based navigation to candidate stations + visual verification at each station + two-layer safety monitor.]

Contributions.

- [TODO: Perception: DINOv3 ViT-S/16 fine-tuned on eight reference images with view-matched augmentation; 32 ms/frame on CPU.]
- [TODO: Planning: grid-based shortest-path planning and nearest-station visiting order.]
- [TODO: Safety: directional proximity monitoring with replanning, plus progress-based stuck detection.]
- [TODO: Evaluation: three student missions plus a 96-run Monte Carlo benchmark with an independent referee.]

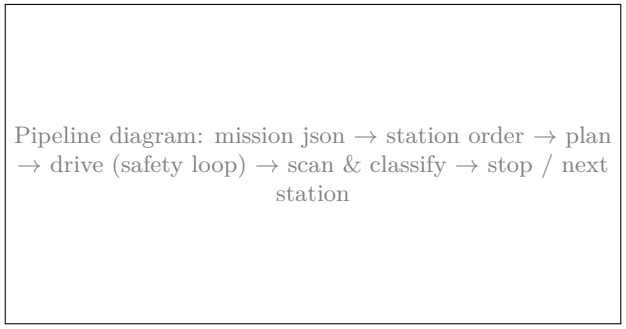

Figure 1. System overview. [TODO: caption]

2. System Overview

[TODO: Mission flow (Fig. 1): read target label; pick nearest unvisited station; plan and drive to its observation cell; turn to observation yaw; classify; stop if match, otherwise continue.]

[TODO: Sensors and what each is used for (camera: target identity only; GPS/IMU + map: localisation and navigation; ps0–ps7: safety). State explicitly what is not used (world files, texture URLs, ground truth).]

[TODO: Code structure: detector/, path/, robot/, safety/, stations/; how the modules are integrated in the main loop.]

3. Perception

[Owner: [TODO: Name]]

3.1. Problem and design rationale

[TODO: Eight known classes, one reference image each → few-shot classification. Constraints: CPU inference, run-time budget, domain gap between clean reference and low, dark, off-axis camera view.]

3.2. Candidate approaches

We compare vision-language models [6] and self-supervised ViTs [2, 7]. [TODO: Compared zero-shot

Method	Normal	Hard	CPU ms
Zero-shot CLIP	[TODO:]	[TODO:]	[TODO:]
Prototype (DINOv3-S)	[TODO:]	[TODO:]	[TODO:]
Fine-tune ResNet18	[TODO:]	[TODO:]	[TODO:]
Fine-tune DINOv3 ViT-S/16	100.0	88.7	32

Table 1. Encoder comparison on captured frames. [TODO: fill from perception report]

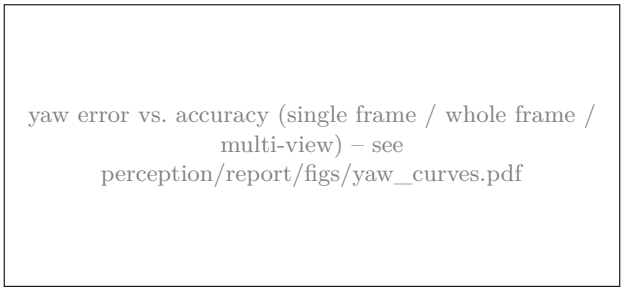

Figure 2. Accuracy against heading error. [TODO: caption]

CLIP/SigLIP2, prototype matching on frozen features (CLIP, DINOv2, DINOv3, ResNet18, MobileNetV3), and full fine-tuning. Why prototypes fail (cosine margin ≈0.02).]

3.3. Method

We fine-tune DINOv3 ViT-S/16 [7] from timm [9] with label smoothing [8]. [TODO: DINOv3 ViT-S/16 (timm), CLS pooling, 224×224 whole-frame resize (no centre crop). View-matched augmentation: lower 40–70% crop of the reference, brightness 0.3–0.8×, dark band below. Label smoothing; fp16 checkpoint (43 MB).]

3.4. Open-set rejection and multi-view scan

[TODO: Confidence threshold 0.20 (distractor FPR@95 = 0%). Scan at each station and keep the most confident frame; yaw error vs. accuracy curve.]

3.5. Perception results

[TODO: Key findings: centre crop loses the poster at yaw error; whole-frame input recovers most of it; auto-crop adds little beyond that; scan reaches ≈100% within ±30°.]

4. Route Planning

[Owner: Isaac Mangano]

4.1. Problem and design rationale

[TODO: The map is fully known, so global planning is exact; the unknown is which station holds the target. Why a grid planner rather than learning-based navigation.]

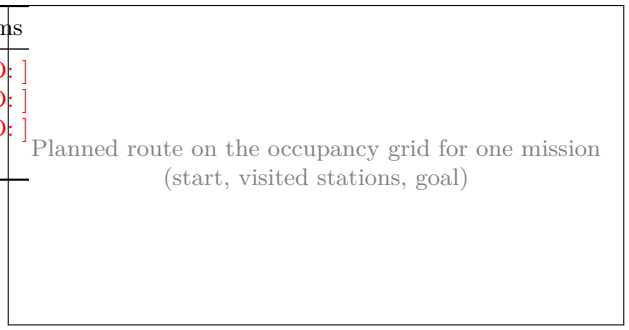

Figure 3. Example plan. [TODO: caption]

4.2. Map representation

[TODO: Occupancy grid from maps/: resolution (0.1 m cells, 40×40), border margin, obstacle clearance, world↔grid conversion.]

4.3. Shortest path

We use Dijkstra's algorithm [1]; A* [3] gives the same path here. [TODO: Dijkstra (uniform cost, 4-connected) from the current cell to the station's observation cell; conversion of the cell path into MOVE / TURN_LEFT / TURN_RIGHT commands. Why not A* (same result on this grid size; mention cost).]

4.4. Station visiting order

[TODO: Greedy nearest unvisited station (Manhattan distance). Discuss alternatives (TSP order, information-driven order).]

4.5. Motion execution

[TODO: Cell-to-cell forward moves, in-place turns to heading, final alignment to the observation yaw, and the station scan (±5°, ±10°).]

5. Safety

[Owner: [TODO: Name (Hyunkyung)]]

5.1. Problem and design rationale

[TODO: Any collision fails the mission. Sources of risk: small clearance in the grid, drift during forward moves, sliding along walls. Why a monitor layered on top of the planner rather than a reactive-only controller.]

5.2. Layer 1: directional proximity monitoring

[TODO: Sensor groups (front ps0/ps7 + diagonals ps1/ps6, sides, back); calibrated thresholds (front 120, right 150, …); per-step check during every move; front

Parameter	Value
Front danger threshold	120
Right danger threshold	150
Stuck window	8 steps
Slow-step ratio	0.75
Max steps per move	75

Table 2. Safety parameters and how they were calibrated. [TODO: complete]

World	Success	Time (s)	Final dist. (m)
Start A	[TODO:]	[TODO:]	[TODO:]
Start B	[TODO:]	[TODO:]	[TODO:]
Start C	[TODO:]	[TODO:]	[TODO:]

Table 3. Results on the three provided student missions.

	Baseline	+ stop fix	Final
Success	0/96	96/96	96/96
Collision / timeout	92 / 4	0 / 0	0 / 0
Time med / p90 / max (s)	102 / – / –	73 / 143 / 167	108 / 197 / 229
Final dist. med / max (m)	1.881 / –	0.110 / 0.157	0.038 / 0.077

Table 4. Monte Carlo benchmark (96 runs per version). [TODO: name the versions; Final = commit 5828c41.]

recovery (short reverse); marking the blocked cell and replanning.]

5.3. Layer 2: progress-based stuck detection

[TODO: Expected 4.0 mm/step at full speed; stuck if none of the last 8 steps exceeds 3.0 mm, or no new cell after 75 steps; calibration from logs (wall sliding 1.8–2.3 mm, head-on ≈0 mm, normal start ≤4 slow steps).]

5.4. Safety results

[TODO: Proximity peak logs per sensor; triggered events in the benchmark; stuck-test scenario (stuck_test controller).]

6. System Evaluation

6.1. Protocol

[TODO: (i) The three provided student missions (worlds/*.wbt): success/failure and completion time. (ii) Monte Carlo benchmark: 3 starts × 8 targets × 4 station layouts (cyclic re-deal of textures) = 96 runs, headless Webots R2025a, scored by an independent supervisor (success = within 0.20 m, no contact above floor level, < 240 s).]

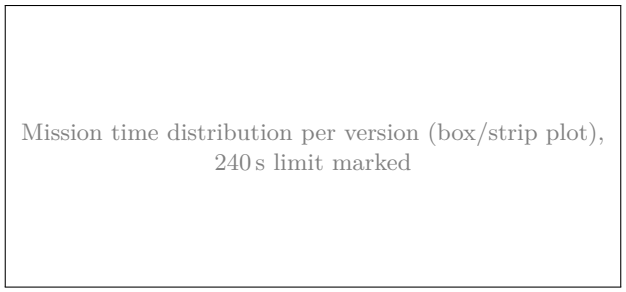

Figure 4. Completion time across the Monte Carlo runs. [TODO: caption]

6.2. Results

[TODO: Discuss Tabs. 3 and 4: effect of the stop fix, precision gained by removing step-back, time cost of the station scan, per-start breakdown.]

6.3. Failure cases

[TODO: Meaningful failures: baseline collisions after reaching the correct station (motors kept reversing after main() returned); backpack/shoe confusion in the early CNN; runs close to the time limit. Each with cause and fix.]

6.4. Ablations

[TODO: Optional: scan vs. single frame; centre crop vs. whole frame; safety layers on/off.]

7. Limitations and Future Work

[TODO: Limitations: time margin (slowest run 229 s of 240 s); scan speed; 4-connected paths with many turns; greedy station order; benchmark uses a fixed set of starts and deterministic physics.]

[TODO: Improvements: faster or confidence-gated scan; turn-cost or 8-connected planning; path smoothing with a pure-pursuit/PID follower; learned local controller (SAC/TD3) compared against PID; randomised start poses and perturbations in the benchmark.]

8. Conclusion

[TODO: 2–3 sentences.]

Contribution Statement

Member	Responsibilities
[TODO: Name]	Perception (encoder study, DINOv3 fine-tuning, detector integration); Monte Carlo benchmark and referee; [TODO: report sections]
Isaac Mangano	Route planning, robot motion, main controller loop and integration; station scan; [TODO:]
[TODO: Name (Hyunkyung)]	Safety monitoring, replanning, stuck detection and calibration; [TODO:]

[TODO: How the work was integrated: shared repository, pull requests and review, Monte Carlo runs on each merged version.]

References

[1] Edsger W. Dijkstra. A note on two problems in connexion with graphs. Numerische Mathematik, 1:269–271, 1959. 2

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[4] Olivier Michel. Webots: Professional mobile robot simulation. International Journal of Advanced Robotic Systems, 1(1):39–42, 2004. 1

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